Memorix Anatomy

14 Endocrine system

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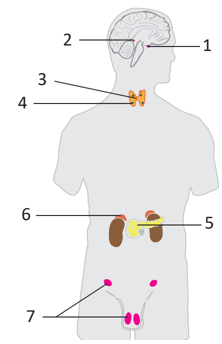
1 General overview

The phylogenetically old endocrine system is defined by its ability to secrete **hormones**, which are chemical messengers that regulate the activity of cells by acting on specific receptors. Both the endocrine system and central nervous system are responsible for controlling and coordinating the functions of organs. However, the endocrine system acts through the secretion of hormones and the CNS by action potentials. Hormone signaling is slower than the transmission of impulses in the nervous system. Therefore, hormones act primarily to regulate long-term metabolic changes and maintain **homeostasis**. The endocrine system consists of **endocrine glands** and the cells of the **diffuse neuroendocrine system (DNES)**. Endocrine glands are also called ductless glands. They secrete hormones directly into the blood stream, rather than through ducts, which is typical for exocrine glands. The diffuse neuroendocrine system is formed by isolated cells dispersed throughout organs not considered to be endocrine glands. Their products are termed tissue hormones and are transported locally by the extracellular fluid, lymph and blood.

Endocrine glands

- 1 **Hypophysis / pituitary gland** (*hypophysis, glandula pituitaria*)
  - 1.1 **Adenohypophysis** (anterior lobe)
  - 1.2 **Neurohypophysis** (posterior lobe)
- 2 **Pineal gland / pineal body** (*glandula pinealis, corpus pineale*)
- 3 **Thyroid gland** (*glandula thyroidea*)
- 4 **Parathyroid gland** (*glandula parathyroidea*) – usually 4 glands
- 5 **Endocrine part of the pancreas** – pancreatic islets of Langerhans
- 6 **Suprarenal/adrenal glands** (*glandulae suprarenales*)
- 7 **Gonads** – ovaries and testes

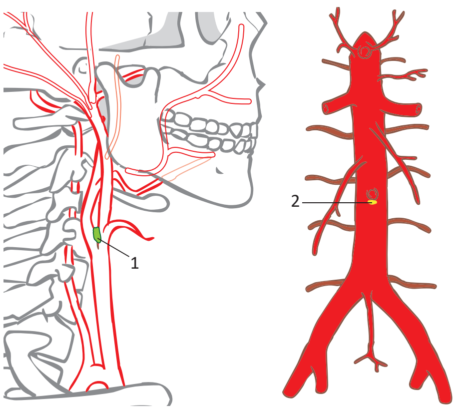


2 Diffuse neuroendocrine system (DNES) and paraganglia

The DNES is formed by **neuroendocrine cells** scattered throughout the epithelium of all organs not considered to be endocrine glands. It is responsible for producing numerous **peptides** that act both as **neurotransmiters and as hormones**. They have a wide variety of functions that include the regulation of intestinal movement and the secretion of digestive enzymes in the digestive system and regulation of the development and function of the respiratory system. **Paraganglia** form a part of the neuroendocrine system. They are formed by clusters of cells originating from the neuroectoderm in the proximity of the **large vessels, sympathetic ganglia and autonomic nerves**.

Paraganglia

- 1 **Carotid body** (*glomus caroticum*)
  - is located at the carotid bifurcation (at level C4)
  - a chemoreceptor
  - contains chromophobe cells
- 2 **Organ of Zuckerkandl** (*paraganglion aorticum abdominale*)
  - is located at the origin of the inferior mesenteric artery
  - consists of chromaffin cells similar to the medulla of the adrenal gland
- 3 **Aortic body** (*glomus aorticus/supracardiacum*)
  - a chemoreceptor located in the aortic arch
- 4 **Coccygeal body / Luschka's body** (*glomus coccygeum*) – is located in front of the coccygeal apex
- 5 **Jugular body** (*glomus jugulare*)
  - is located in the superior bulb of the internal jugular vein
- 6 **Tympanic body** (*glomus tympanicum*)
  - is situated within the wall of the tympanic cavity



Paraganglia in proximity to large arteries

The hypophysis is attached to a piece of dura mater called the sellar diaphragm *(diaphragma sellae)* at the hypophysial fossa *(sella turcica)*. The sellar diaphragm contains an opening (the foramen of Pacchioni) for the pituitary stalk.

Herring's bodies *(corpuscula neurosecretoria)* are synonyms for the secretory granules in the hypothalamohypophysial tract.

The adenohypophysis is derived from Rathke's pouch, which is a pouch of ectoderm originating from the roof of the stomodeum in the third week of development. Small cavities (Rathke's follicles) may persist between the distal and intermediate parts as remnants of this pouch. A tumour called a craniopharyngioma can emerge in this area.

The carotid sinus is the enlarged beginning of the internal carotid artery. Baroreceptors that register changes in the blood pressure are located within the wall of the carotid sinus.

Testis see page 240.
Ovary see page 250.
Hypothalamus see page 430.

Clinical notes

Carcinoid tumors originate from the cells of the gastrointestinal diffuse neuroendocrine system. They are one of the most common neuroendocrine tumors of the gastrointestinal tract. Carcinoids often produce serotonin and histamine, the overproduction of which leads to the carcinoid syndrome, which is characterised by flushing, diarrhoea, bronchoconstriction and cardiac changes.

Massage of the carotid sinuses can lead to a decrease in heart rate via activation of the vagus nerve. This can be used as the first-line treatment of tachycardia.

Adenohypophyseal tumours can manifest with bitemporal heteronymous hemianopsia as a result of compression of the optic chiasm. Pituitary surgery can be performed through the nasal cavity and sphenoidal sinus. This is termed transsphenoidal surgery. The traditional surgical approach can be replaced by radiotherapy with Leksell´s gamma knife.

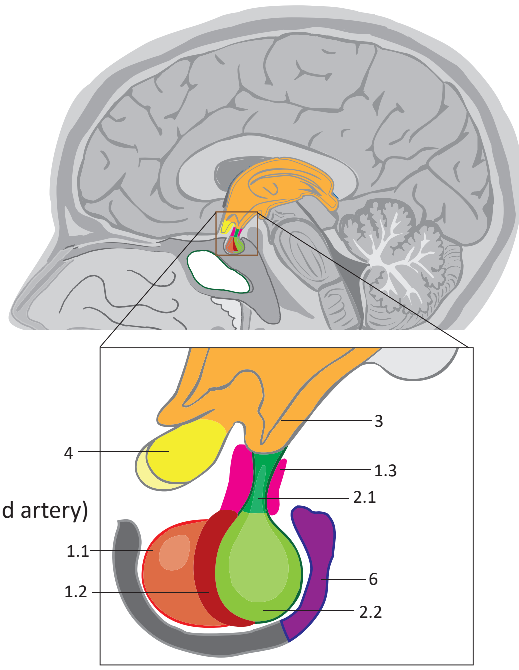
Craniopharyngeoma is a tumour derived from the residual embryonic tissue of Rathke's pouch. It can manifest as a functional lesion of the pituitary gland or as a visual field impairment from compression of the optic chiasm.

The pituitary gland is a small gland located in the **hypophyseal fossa** of the **sphenoid bone** *(sella turcica)*. The pituitary stalk connects the pituitary gland to the hypothalamus. The hypothalamus controls the activity of the pituitary gland. The hypophysis and hypothalamus form the **hypothalamohypophysial system/axis,** which is responsible for the majority of hormonal regulation and is also closely related to **the autonomic nervous system**. The anterior part of the pituitary gland is called **adenohypophysis** and is derived from the roof of the stomodeum (ectoderm). The posterior part, termed **neurohypophysis,** represents a pouch of the diencephalon at the base of the third ventricle (derived from the neuroectoderm).

Structure

1 **Adenohypophysis (anterior lobe)**
- 1.1 **Pars distalis** – the major part of the anterior lobe
- 1.2 **Pars intermedia** – a strip of hypophyseal tissue at the border with the posterior lobe
- 1.3 **Pars tuberalis** – a layer surrounding the stalk of the posterior lobe

2 **Neurohypophysis (posterior lobe)**
- 2.1 **Infundibulum** – the stalk of the posterior lobe
- 2.2 **Neural lobe** – the posterior lobe itself



Close-up of a sagittal section of hypothalamus and hypophysis

Syntopy

- 3 **Cranially:** hypothalamus (median eminence, tuber cinereum)
- 4 **Ventrocranially:** optic chiasm
- 5 **Ventrocaudally:** sphenoid sinus
- 6 **Dorsally:** dorsum sellae, basilar artery and pons
- 7 **Laterally:** cavernous sinus (with 3^{rd}, 4^{th}, 5^{th} and 6^{th} cranial nerves and internal carotid artery)

Histological structure

Anterior lobe (adenohypophysis)

1 **Pars distalis**
- 1.1 **Chromophobe cells** – stem cells or hormone secreting cells
- 1.2 **Chromaffin cells** – somatotrophs, mammotrophs, gonadotrophs, thyreotrophs, corticotrophs (named according to the target gland)

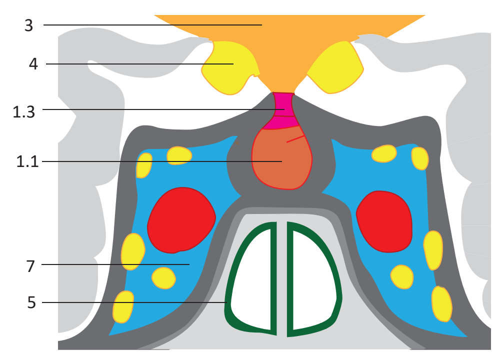
2 **Pars intermedia** – corticotrophs

3 **Pars tuberalis** – gonadotrophs or thyreotrophs

Posterior lobe (neurohypophysis)

1 **Axons of neuroendocrine neurons** – the cell bodies are positioned in the supraoptic and paraventricular nuclei of the hypothalamus
– the hypothalamohypophysial tract transports secretory granules to the hypophysis

2 **Pituicytes** – glial cells



Frontal section of the hypophysis and cavernous sinus

Functional connections with the hypothalamus

Anterior lobe (adenohypophysis)

1. hormones are secreted in **the arcuate nucleus** (parvocellular part) of the hypothalamus and stimulate or inhibit cells of the adenohypophysis
2. hormones are carried via axonal transport to **the median eminence** where there are released into capillaries during a process called neurosecretion
3. the **portal system of the hypophysis transmits hormones** to the chromaffin cells
4. **secretion of hormones into the blood stream**

Posterior lobe (neurohypophysis)

1. **neuroendocrine secretion from axons of the hypothalamohypophysial tract** (supraoptic and paraventricular nucleus) to the drainage area of the inferior hypophysial artery

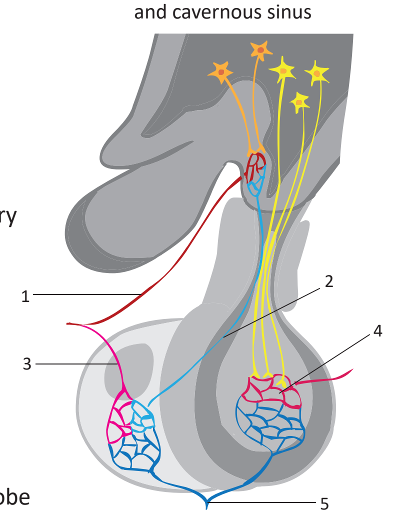
Blood supply

Arterial blood supply:

- 1 **Superior hypophysial artery** (a branch of the internal carotid artery) – supplies the infundibulum and median eminence
- 2 **Portal system of the hypophysis** – originates from the spiral arteries in the pituitary stalk and continues to the second capillary bed in the adenohypophysis
- 3 **Trabecular artery** – supplies the pars distalis of the anterior lobe
- 4 **Inferior hypophysial artery** (branch of the internal carotid artery) – supplies the posterior lobe

Venous drainage:

- 5 **Hypophysial veins** (into the cavernous sinus)



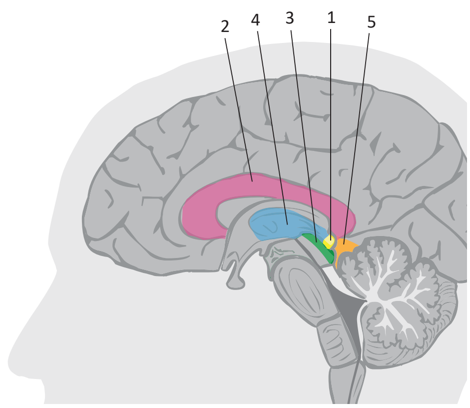
Scheme of the hypothalamohypophysial system (sagittal section)

4 **Pineal gland** – Glandula pinealis / Corpus pineale

The pineal gland is an unpaired oval-shaped organ. It is part of epithalamus, which is located the diencephalon. The pineal gland is connected to the habenular commisure by its stalk and is positioned in the quadrigeminal cistern rostral the superior colliculi of the midbrain. **Melatonin, the pineal hormone, is secreted primarily in darkness** and is involved in **the control of circadian rhythm**. Information about light from the retina is transmitted to the pineal gland by a branch of the visual pathway that passes through the hypothalamus.

Syntopy

- 1 **Pineal gland**
- 2 **Cranially:** corpus callosum
- 3 **Caudally:** quadrigeminal plate
- 4 **Rostrally:** third ventricle
- 5 **Dorsolaterally:** quadrigeminal cistern
- 6 **Dorsally:** great cerebral vein



Sagittal section of the brain

Histological structure

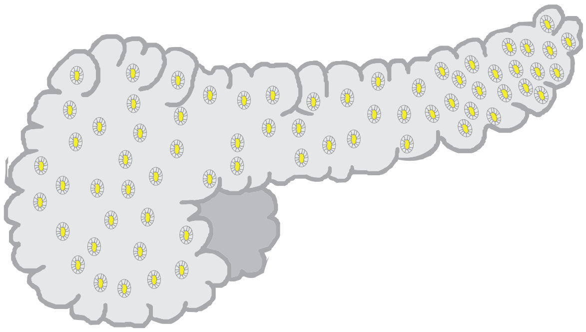
1 **Pia mater** – is located on the surface of the gland
– emits septa inside the gland, which contain vessels and unmyelinated nerve fibres
2 **Pinealocytes** – the melatonin secreting cells
3 **Interstitial cells** – specific astrocytes of the pineal gland

5 Endocrine component of pancreas– pancreatic islets

The endocrine component of the pancreas is composed of clusters of endocrine cells termed **islets of Langerhans**. The islets of Langerhans **are scattered throughout the pancreas** with the highest density in the pancreatic tail. The islets of Langerhans are richly vascularised by fenestrated capillaries and are covered by fibrous capsules. **The endocrine part of the pancreas is responsible predominantly for insulin and glucagon production and for controlling the concentration of glucose in the blood (glycaemia).**

Histological structure

A-cells – are located in the periphery of the islets
– produce glucagon
B-cells – constitute the main component of the islets
– produce insulin
D-cells – the minority of cells
– are located in the periphery of the islets and produce somatostatin
D1 cells – produce a peptide similar to vasoactive intestinal peptide
PP-cells – produce vasoactive intestinal peptide (VIP) and pancreatic peptide (PP)
EC-cells – produce substance P and serotonin


Pancreas with the islets of Langerhans

Phylogenetically, the pineal gland represents a remnant of the neuroepiphysis. In some vertebrates, the neuroepiphysis is composed of the pineal gland and the parapineal body situated at the surface of the head and serves as a receptor for light intensity and the day-night cycle. Thus, it is called the parietal eye or pineal eye.

The islets of Langerhans develop in the third month of intrauterine development and insulin secretion begins in the fifth month. The islet diameter is approximately 0.1–0.5 mm and the total number of islets is about one million. The islets of Langerhans constitute 2–3 % of the total weight of the pancreas.

Pyramidal lobe / Lalouette's pyramid *(lobus pyramidalis)* is present in 25–50 % of cases.

Tubercle of Zuckerkandl *(tuberculum dorsale)* is the most dorsal structure of the thyroid lobes.

Thyroid ima artery / thyroid artery of Neubauer *(arteria thyroidea ima)* is present in 2 % of cases. It commonly extends from the aortic arch and supplies the inferior part of the thyroid gland. There is a risk of injury to this artery during tracheostomy.

Clinical notes

Brain sand *(acervulus cerebri)* is a term for calcium deposits in the pineal gland. It can be used as an anatomical landmark in radiological studies.

Pineal gland cysts can compress the cerebral aqueduct, resulting in hydrocephalus.

Diabetes mellitus type 1 occurs as a result of the autoimmune destruction of B-cells mediated by autoantibodies. It is possible to transplant the islets of Langerhans. An emulsion is injected (instilled) into the portal vein and the islets are implanted in the liver.

Goitre *(struma)* is a term for an enlarged thyroid gland.

During thyroidectomy, the surgical procedure to resect the thyroid gland, the recurrent laryngeal nerve can be harmed and the parathyroid glands can be accidentally removed. Injury of the recurrent laryngeal nerve leads to dysphonia, an impairment in voice production. Removal of all parathyroid glands can manifest as severe and life-threatening hypocalcaemia.

The thyroid gland is composed of two lobes connected by an isthmus. The thyroid gland is situated **in front of the trachea** and the isthmus is located **at the level of the second to fourth tracheal rings**. However, in some cases, the thyroid can be located behind the manubrium of the sternum. **Two parathyroid glands are usually positioned on the medial side of each lobe.** Howevever, the location and number of the parathyroid glands can vary. **The gland is supported by thickened strips of the surrounding fibrous tissue**, especially around the vessels. **The thyroid gland secretes the hormones triiodothyronine, thyroxine (tetraiodothyronine), calcitonin and pro-calcitonine. The parathyroid glands produce parathormone.**

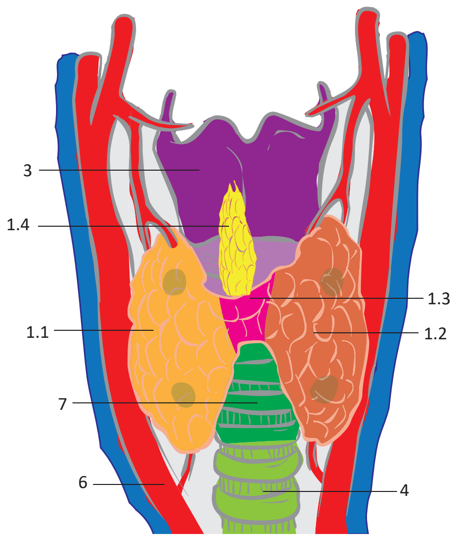
Structure

1 **Thyroid gland** (*glandula thyroidea*)
- 1.1 **Right lobe** (*lobus dexter*)
- 1.2 **Left lobe** (*lobus sinister*)
- 1.3 **Isthmus**
- 1.4 **Pyramidal lobe** (*lobus pyramidalis*)

- 2 **Parathyroid glands** (*glandulae parathyroidae*)

Syntopy

- 3 **Cranially:** cricothyroid muscle and thyroid gland
- 4 **Caudally:** trachea, unpaired thyroid plexus, brachiocephalic veins
- 5 **Ventrally:** infrahyoid muscles
- 6 **Laterally:** internal carotid artery, internal jugular vein
- 7 **Medially from lobes:** the inferior part of the larynx, trachea
- 8 **Dorsomedially from lobes:** recurrent laryngeal nerve (in the tracheooesophageal sulcus)

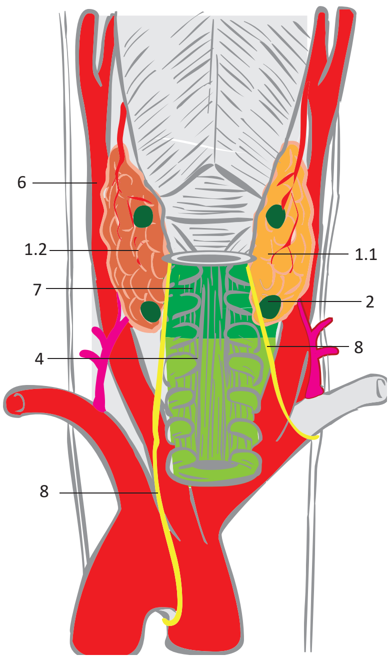


Thyroid gland, anterior view

Histological structure

Thyroid gland
1 **Capsule** – is divided into two sheets, the venous plexus is situated between them
 1.1 **Capsula propria** – is located on the surface of the gland and emits fibrous septa
 1.2 **Capsula externa** – is continuous with the pretracheal layer of the cervical fascia
2 **Lobules** – contain the colloid-filled thyroid follicles
3 **Follicular cells** – produce triiodothyronine (T3) and thyroxine (T4) hormones
4 **Parafollicular cells (C cells)** – produce calcitonin

Parathyroid glands
1 **Fibrous capsule**
2 **Parenchymal tissue**
3 **Chief cells** – produce parathormone
4 **Oxyphil cells**



Thyroid gland, posterior view

Blood supply

Arterial blood supply:
– **thyroid gland:** superior thyroid artery (from the external carotid artery)
– **thyroid and parathyroid gland:** inferior thyroid artery (from the thyrocervical trunk)

Venous drainage:
– superior thyroid vein (to the internal jugular vein)
– middle thyroid vein (to the internal jugular vein)
– inferior thyroid vein, veins on both sides are form the unpaired thyroid plexus (to the left brachiocephalic vein)

Lymphatic drainage:
– subcapsular plexus (to the deep cervical lymph nodes)

Innervation

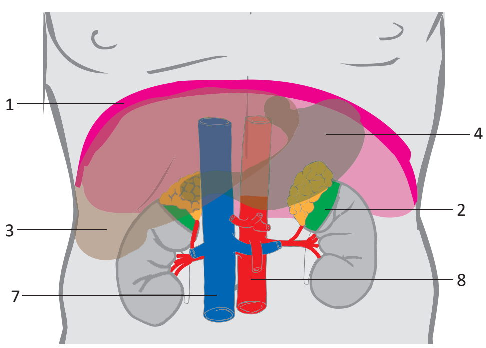
Sympathetic innervation:
– cervical ganglia (nerves entering the gland along arteries)

Parasympathetic innervation:
– **thyroid gland:** superior and recurrent laryngeal nerve (both from the vagus nerve)
– **parathyroid gland:** recurrent laryngeal nerve (from the vagus nerve)

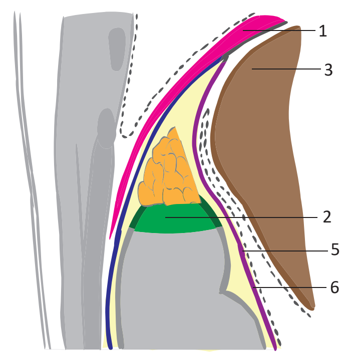
The adrenal gland is a paired organ. The right adrenal gland is triangular in shape and the left adrenal gland is semilunar-shaped. **The adrenal glands are positioned on the upper poles of the kidneys (retroperitoneally) at the level of T11–T12**. The right adrenal gland is situated more caudally than the left one. **Fixation of the adrenal glands is provided by vessels**. The adrenal veins leave the glands through the hilum, which is situated on the anterior surface. The adrenal gland is covered by a thin capsule which emits septa into the gland. On cross section, the gland consists of cortex and medulla. **The cortex produces mineralocorticoids, glucocorticoids and androgens. The medulla is responsible for catecholamine secretion.**

Syntopy

- 1 **Cranially:** diaphragm, subhepatic space
- 2 **Caudally:** upper pole of the kidney
- 3 **Ventrally on the right side:** liver
- 4 **Ventrally on the left side:** stomach
- 5 **Ventrally on both sides:** prerenal layer of the renal fascia, perirenal fat capsule
- 6 **Dorsally:** perirenal fat capsule, retrorenal layer of the renal fascia, pararenal fat body, diaphragm, costodiaphragmatic recess
- 7 **Medially on the right side:** inferior vena cava
- 8 **Medially on the left side:** abdominal aorta
- 9 **Laterally:** pararenal fat body



Syntopy of the adrenal glands, anterior view

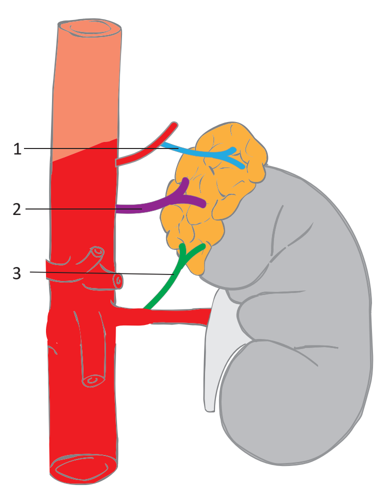


Syntopy of the right adrenal gland in the sagittal section of the body

Histological structure

Cortex
1 **Zona glomerulosa** – groups of polyedric cells
– produces mineralocorticoids
2 **Zona fasciculata** – parallel fascicles
– produces glucocorticoids
3 **Zona reticularis** – spatial arrangement of fascicles
– produces androgens

Medulla
1 **A cells** – produce epinephrine
2 **N cells** – produce norepinephrine
3 **Ganglion cells**



Anterior view of the aorta, left kidney and adrenal gland

Blood supply and innervation

Arterial blood supply:
- 1 **Superior adrenal/suprarenal artery** – from the inferior phrenic artery
- 2 **Middle adrenal/suprarenal artery** – from the abdominal aorta
- 3 **Inferior adrenal/suprarenal artery** – from the renal artery

Venous drainage:
– suprarenal veins (on the right side into the inferior vena cava, on the left into the left renal vein)

Lymphatic drainage: lumbar nodes and posterior mediastinal nodes

Sympathetic and parasympathetic part of the autonomic nervous system, sensory:
– adrenal/suprarenal plexus

The accessory suprarenal glands / Marchand´s adrenals can be present anywhere along the path of the descent of the gonads.

Cortical and medullar arteries originate from the subcapsular plexus (formed by the suprarenal arteries). Veins are not present in the cortex and blood is drained into the capillaries of the medulla. This arrangement is important for the regulation of epinephrine secretion.

Fascia of Toldt is an eponym for the prerenal layer of the renal fascia (Gerota's fascia).

Fascia of Zuckerkandl is the retrorenal layer of the renal fascia (Gerota's fascia).

Clinical notes

Phaeochromocytoma is a tumour of the adrenal medulla that causes an irregular secretion of catecholamines, which manifests as paroxysms of hypertension and tachycardia .

Cushing syndrome is a term for the collection of symptoms caused by high levels of glucocorticoids. This syndrome is most commonly caused by iatrogenic administration of cortisol but may also be a manifestation of a pituitary adenoma.

Conn's syndrome is a term for a hyper-functioning adrenal cortex causing an increased production of mineralocorticoids (hyperaldosteronism).

Addison's disease is a term for a hypo-functioning adrenal cortex with insufficient production of glucocorticoids (hypocorticalism).

Tissue or organ	Hormones
Brain	endorphins, enkephalins, dynorphin, serotonin
Hypothalamus	thyrotropin-releasing hormone, gonadotropin-releasing hormone, growth hormone-releasing hormone, corticotropin-releasing hormone, somatostatin, dopamine
Adenohypophysis	somatotropin, prolactin, adrenocorticotropic hormone, thyroid-stimulating hormone, luteinizing hormone, follicle-stimulating hormone
Neurohypophysis	oxytocin, vasopressin
Epiphysis	melatonin
Thyroid gland	triiodothyronine, thyroxine (tetraiodothyronine), calcitonin
Parathyroid gland	parathormone
Heart	atrial natriuretic factor
Stomach, intestine	gastrin, ghrelin, cholecystokinin, IGF, motilin, neuropeptide Y, somatostatin, serotonin
Liver	IGF, thrombopoietin
Islets of Langerhans	glucagon, insulin, somatostatin, pancreatic polypeptide
Kidneys	calcitriol (the active form of vitamin D3), erythropoietin, serotonin
Adrenal cortex	aldosterone, cortisol, dihydroepiandrosterone, androstenedione
Adrenal medulla	epinephrine, norepinephrine, dopamine
Testes	testosterone
Ovarian follicle	estrogens (estradiol, estrone, estriol)
Corpus luteum	gestagens (progesterone), estrogen (estradiol, estrone, estriol), androstenedione
Placenta	gestagens (progesterone), estrogen (estradiol, estrone, estriol), human chorionic gonadotropin (hCG), human placental lactogen / chorionic somatomamotropin (HCP hPL), hACTH
Skin	calciferol, cholecalciferol (provitamin D)
Adipose tissue	leptin, adiponectin

9 Review questions and figures

I. General part

1. Explain the pathway of hormones from their place of origin to their targeted structures. (p. 522)
2. Name the seven endocrine glands. (p. 522)

II. Diffuse neuroendocrine system

3. Define the term paraganglia. (p. 522)
4. Explain the relationship between the carotid body and carotid sinus (p. 522)
5. Define the location of the Organ of Zuckerkandl. (p. 522)

III. Pituitary Gland

6. Describe the structure of the pituitary gland. Name the two lobes and their parts. (p. 523)
7. Name the structures surrounding the hypophysis. (p. 523)
8. Name the hormones produced in the adenohypophysis. (p. 523)
9. Explain the term "portal system of the hypophysis" and state its function. (p. 523)

IV. Pineal Gland

10. Describe the position of the pineal gland and the structures in its proximity. (p. 524)
11. Name the hormone/s produced in the pineal gland. (p. 524)

V. Endocrine components of the pancreas

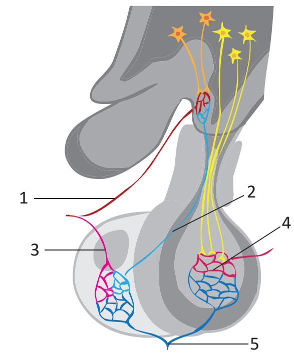
12. Describe the location of the islets of Langerhans in the pancreas. (p. 524)
13. Name the cell types in the islets of Langerhans. (p. 524)

VI. Thyroid and parathyroid gland

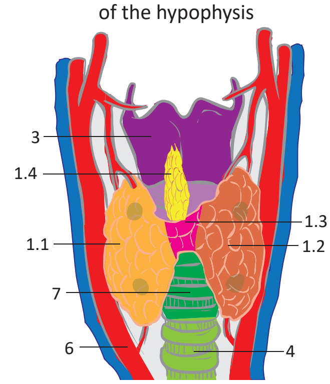
14. Name the parts of the thyroid gland. (p. 525)
15. Draw the relationship of the thyroid gland to the recurrent laryngeal nerve. (p. 525)
16. Name the arteries and veins of the thyroid gland and describe the blood supply of the parathyroid glands. (p. 525)
17. Specify the cell types of the thyroid gland and name the secreted hormones. (p. 525)

VII. Adrenal gland

18. Name all the layers of the adrenal cortex. (p. 526)
19. Name the arteries supplying the adrenal gland. (p. 526)



Describe the blood supply of the hypophysis



Describe the structure and syntopy of the thyroid gland

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